**1. Project Goal**
Build a standalone handheld FPGA computer using a custom 8-bit CPU, custom instruction set, display, buttons, PCB and enclosure.

**2. CPU Architecture**
Data width: 8-bit
Address width: 16-bit
Registers: 4 general-purpose registers
Instruction width: 16-bit
ALU: 8-bit
Architecture: custom CPU implemented in FPGA

**3. Registers**
R0 - General-purpose 8-bit register
R1 - General-purpose 8-bit register
R2 - General-purpose 8-bit register
R3 - General-purpose 8-bit register

PC - 16-bit Program Counter
Stores the address of the next instruction.

SP - 16-bit Stack Pointer
Points to the current location of the stack in memory.

FLAGS - Status register
Stores:
Z - Zero flag
C - Carry flag
N - Negative flag

**4. Instruction Set**
LOAD Load a value from memory into a register
STORE Store a register value into memory
MOV Copy a value between registers
ADD Add two values
SUB Subtract two values
AND Bitwise AND
OR Bitwise OR
XOR Bitwise XOR
CMP Compare two values and update flags
JMP Jump to another instruction
JZ Jump if Zero flag is set
JNZ Jump if Zero flag is not set
PUSH Push a value onto the stack

POP  Remove a value from the stack
IN  Read input from an input device
OUT  Send data to an output device
HALT  Stop CPU execution
LOADI  Load an immediate value into a register

**5. Memory Model**
ByteForge uses separate program memory and data memory.

Program memory stores CPU instructions and is addressed by the 16-bit Program Counter.

The data-memory address space is 16-bit, allowing up to 65,536 addressable locations. This space contains RAM and memory-mapped input/output devices.

The stack is stored inside RAM. Certain addresses in the data-memory space will be reserved for hardware such as the buttons and display.

**6. Input/Output**
Inputs:
- Directional buttons
- Action buttons
- Possibly additional system buttons

Outputs:
- Small colour display
The CPU will communicate with the input and output hardware using memory-mapped I/O.
Reading from specific addresses will allow the CPU to determine which buttons are pressed.
Writing to specific addresses will allow the CPU to control the display.

**7. Final Success Criteria**
ByteForge V1 will be considered successful if:
1. The custom 8-bit CPU runs correctly on an FPGA.
2. The CPU can execute programs using the custom instruction set.
3. Programs can read physical button inputs.
4. Programs can output graphics or text to the display.
5. The CPU and supporting hardware are integrated onto a custom PCB.
6. The device is powered and programmable through an appropriate external connection such as USB-C.
7. The electronics fit inside a custom enclosure.
8. A simple assembler can convert ByteForge assembly code into machine code.
9. The finished computer can run multiple demonstration programs, including:
   - an arithmetic/calculator demonstration
   - an interactive input/output demonstration
   - a simple game such as Pong or Snake

**8. Instruction Encoding**

ByteForge uses 16-bit instruction words.

The first 5 bits are used for the opcode. This gives up to 32 possible instruction types, which is enough for the current instruction set while leaving room for future expansion.

Because ByteForge has four general-purpose registers, each register can be identified using 2 bits:

00 = R0

01 = R1

10 = R2

11 = R3

Different instructions use the remaining bits differently depending on whether they operate on registers, immediate values, or memory addresses.

For register-based instructions such as ADD R1, R2, the instruction contains the opcode and the two register selections in the same 16-bit word.

For immediate instructions such as LOADI R0, 25, the instruction contains the opcode, destination register and an 8-bit constant value.

Instructions that need a full 16-bit memory address use a second 16-bit word immediately after the main instruction. For example:

LOAD R1, [0x2000]

would use one word for the LOAD instruction and register selection, followed by a second word containing the address 0x2000.

OPCODE TABLE

| BINARY | INSTRUCTION | PURPOSE |
|---|---|---|
| 00000 | LOAD | Load a value from memory into a register |
| 00001 | STORE | Store a register value into memory |
| 00010 | MOV | Copy a value between registers |

| | | |
|---|---|---|
| 00011 | LOADI | Load an immediate value into a register |
| 00100 | ADD | Add two register values |
| 00101 | SUB | Subtract one register value from another |
| 00110 | AND | Perform bitwise AND |
| 00111 | OR | Perform bitwise OR |
| 01000 | XOR | Perform bitwise XOR |
| 01001 | CMP | Compare two register values and update flags |
| 01010 | JMP | Jump to another instruction |
| 01011 | JZ | Jump if the Zero flag is set |
| 01100 | JNZ | Jump if the Zero flag is not set |
| 01101 | PUSH | Push a register value onto the stack |
| 01110 | POP | Pop a value from the stack into a register |
| 01111 | IN | Read data from an input device |
| 10000 | OUT | Send data to an output device |
| 10001 | HALT | Stop CPU execution |

The remaining opcode values are left unused so more instructions can be added later if needed.

## 9. CPU Datapath

The CPU datapath describes how data moves between the main parts of the ByteForge processor.

The Program Counter stores the address of the next instruction to be executed.

That address is sent to Program Memory, which returns the instruction stored at that location.

The instruction is placed into the Instruction Register, then passed to the Instruction Decoder.

The Instruction Decoder reads the opcode and register fields. The Control Unit then determines which parts of the CPU need to be activated.

The four general-purpose registers, R0 to R3, can provide values to the 8-bit ALU.

The ALU performs arithmetic and logic operations such as addition, subtraction, AND, OR and XOR. The result can then be written back into one of the registers.

The ALU also updates the FLAGS register, including the Zero, Carry and Negative flags.

The CPU communicates with RAM and memory-mapped input/output devices through the Memory Interface.

This allows programs to read physical button inputs and send data to the display.

**10. Fetch-Decode-Execute Cycle**

ByteForge will execute instructions using a fetch-decode-execute cycle.

Fetch:
The Program Counter contains the address of the next instruction. This address is sent to Program Memory, and the instruction is loaded into the Instruction Register.

Decode:
The Instruction Decoder reads the opcode and register fields. The Control Unit determines what operation is required and which parts of the CPU need to be activated.

Execute:
The CPU carries out the instruction. This could involve performing an ALU operation, reading or writing memory, moving data between registers, accessing an input/output device, or changing the Program Counter for a jump instruction.

After the instruction is completed, the Program Counter moves to the next instruction unless the current instruction changes it.

**11. Memory Architecture Refinement**

ByteForge will use separate program memory and data memory.

Program memory stores the instructions executed by the CPU and is addressed using the Program Counter.

Data memory contains RAM and memory-mapped input/output devices.

The stack is stored inside RAM and is controlled by the 16-bit Stack Pointer.

Separating program memory from data memory simplifies the CPU design because instruction fetching and data access can be handled independently.

**12. Stack Operation**

The Stack Pointer stores the address of the current top of the stack.

The stack is stored inside RAM.

PUSH:

1. The Stack Pointer is decremented.

2. The selected register value is written to the RAM address pointed to by the Stack Pointer.

POP:

1. The value at the RAM address pointed to by the Stack Pointer is loaded into the selected register.

2. The Stack Pointer is incremented.

The stack provides temporary storage for programs and can later support subroutines if CALL and RET instructions are added in a future version.

**13. ALU Implementation and Testing**

ByteForge's 8-bit ALU has been implemented in Verilog. It takes two 8-bit inputs and a 5-bit opcode, and produces an 8-bit result with Zero, Carry and Negative flag outputs. The ALU is combinational and does not require a clock.

It supports ADD, SUB, AND, OR, XOR and CMP. CMP produces the same subtraction result and flags as SUB; the CPU control unit will later prevent that result from being written into a register.

The Zero flag is set when the result is zero. The Negative flag follows bit 7 of the result. For ADD, Carry indicates that the unsigned sum exceeds 255. For SUB and CMP, Carry indicates an unsigned borrow, when A is less than B. Logic operations clear Carry. Unsupported opcodes produce a zero result with all flags cleared.

The ALU was simulated using Icarus Verilog on EDA Playground. All 38 test cases passed with zero failures, covering arithmetic, bitwise operations, flag behaviour and unsupported opcodes.

Simulation: https://www.edaplayground.com/x/R8vK

**14. Register File, FLAGS and Reset**

ByteForge's register file and FLAGS storage have been implemented in Verilog.

The register file contains four writable 8-bit registers, R0–R3. Two combinational read ports allow two register values to be read simultaneously without waiting for a clock edge. A single write port updates the selected register on the rising clock edge when write_enable is set. Otherwise, the registers retain their stored values.

The FLAGS module stores the Zero, Carry and Negative flags. These update on the rising clock edge when flags_write_enable is set; otherwise, their previous values are preserved. Register writes and flag updates have independent enables, allowing them to occur separately or together.

Both modules use an active-high synchronous reset. When reset is asserted, the next rising clock edge clears all four registers and the three stored flags to zero. Reset takes priority over writing.

The modules were tested using a self-checking testbench in Icarus Verilog on EDA Playground. All 131 checks passed with zero failures. The checks covered register reads and writes, value retention, independent flag updates, simultaneous updates, and synchronous reset behaviour.

Simulation: https://www.edaplayground.com/x/ADVd

**15. Instruction Encoding and Decoder Rules**

ByteForge uses 16-bit instruction words. The opcode occupies bits 15–11. Register selections use two bits: 00 = R0, 01 = R1, 10 = R2, and 11 = R3.

Instruction formats

| Format | Bits 15-11 | Bits 10-9 | Bits 8-7 | Remaining bits |
|---|---|---|---|---|
| Two Register | Opcode | Register A | Register B | Bits 6-0 reserved |
| Immediate | Opcode | Destination register | Part of immediate | Immediate occupies bits 8-1; bit 0 reserved |
| Single register | Opcode | Selected Register | Reserved | Bits 8-0 reserved |
| Address only | Opcode | Reserved | Reserved | Bits 10-0 reserved |
| HALT | Opcode | Reserved | Reserved | Bits 10-0 reserved |

All reserved bits must be zero in valid ByteForge V1 instructions.

Instruction groups and lengths

| Instructions | Format | Length |
|---|---|---|
| MOV, ADD, SUB, AND, OR, XOR, CMP | Two register | 1 word |
| LOADI | Immediate | 1 word |
| PUSH, POP | Single register | 1 word |
| LOAD, STORE, IN, OUT | Single register followed by a 16 bit address | 2 words |
| JMP, JZ, JNZ | Address only followed by a 16 bit target | 2 words |
| HALT | HALT | 1 word |

The opcode values remain those defined in Section 8.

Register and operand meaning

For MOV, Register A is the destination and Register B is the source. For arithmetic and logic operations, Register A supplies the first operand and receives the result; Register B supplies the second operand. SUB calculates A − B.

CMP calculates A − B and updates flags without changing either register.

LOAD, LOADI, IN and POP write to the selected register. STORE, OUT and PUSH read from the selected register.

IN and OUT use the same data-memory address space as LOAD and STORE, allowing access to memory-mapped devices.

Program addressing

Program memory is addressed in 16-bit words. A one-word instruction advances the Program Counter by one; a two-word instruction advances it by two, unless a taken jump replaces the Program Counter.

The second word contains the complete address or jump target. Jump targets are program-word addresses; LOAD, STORE, IN and OUT addresses refer to 8-bit locations in data memory.

Decoder behaviour

The decoder is combinational and requires no clock or reset. It extracts the opcode, relevant register selections and immediate value, and indicates whether a second address word is required.

It marks unsupported opcodes or nonzero reserved bits as invalid. Invalid instructions must not request register writes, flag updates, memory operations, jumps or HALT.

The decoder identifies the instruction; the later control unit handles fetching the second word and executing the operation.

**16. Instruction Decoder Implementation and Testing**

ByteForge's instruction decoder was implemented in Verilog according to the encoding rules in Section 15. It reads the first 16-bit instruction word, identifies the opcode, extracts register selections and immediate values, and indicates whether a second word is required.

The decoder produces intent signals for register writes, flag updates, memory reads and writes, jumps, stack operations and HALT. These signals describe the actions required by an instruction. A future clocked control unit will determine when those actions occur.

Unsupported opcodes and instructions with nonzero reserved bits are marked invalid. For invalid instructions, all decoded fields and intent signals are cleared, while the raw opcode remains available. The decoder is combinational, with default assignments on every path to prevent inferred latches.

The self-checking testbench was run using Icarus Verilog in EDA Playground. It completed 1,781 checks with zero failures. Testing covered all 18 supported opcodes, every two-register combination, all 256 immediate values for each destination register, single-register selections, jump conditions, unsupported opcodes, reserved-bit violations and transitions between valid and invalid instructions.

The simulation generated instruction_decoder_waveform.vcd. The source files are instruction_decoder.v and instruction_decoder_tb.v, and the results screenshot is saved as ByteForge_Decoder_Test_Results.png.

Saved simulation: ByteForge Instruction Decoder Tests.

## 17. Control Unit and Instruction Execution Rules

ByteForge will use a clocked control unit to coordinate instruction fetching, decoding and execution. The control unit will convert the decoder's intent signals into timed actions for the ALU, registers, FLAGS, program memory, data memory, Program Counter and Stack Pointer. Only one instruction will execute at a time.

Clock and reset

State changes and stored-value updates occur on the rising clock edge. Reset is active-high and synchronous, taking priority over execution. Reset clears the Program Counter to 0x0000, clears the registers and FLAGS, cancels pending CPU requests, and returns the control unit to instruction fetching.

The Stack Pointer resets to a configurable STACK_TOP value. The exact value and permitted stack address range will be fixed when the physical RAM capacity and final memory map are selected, before CPU integration.

Memory interface and timing

Program and data memory use a request/ready interface. The CPU asserts a request and holds the address, operation and write data stable until memory asserts ready. A transaction completes on a rising clock edge when both request and ready are high.

For reads, memory must provide valid read data at that completion edge. For writes, memory must perform the write exactly once at that edge. The CPU then leaves the waiting state and removes the request. This allows memory operations to take one or more clock cycles.

While waiting for memory, the CPU does not repeat register writes, flag updates or Stack Pointer changes.

Instruction fetching

The control unit requests the 16-bit program word at the Program Counter. When the read completes, it stores the word in the Instruction Register and increments the Program Counter by one.

The instruction is then decoded. If it requires a second word, the CPU requests the word at the updated Program Counter, stores it in an Address Register, and increments the Program Counter again.

Consequently, before execution, the Program Counter points to the next sequential instruction. A taken jump replaces this value with its target address.

Register and ALU instructions

MOV copies Register B into Register A. LOADI writes the decoded 8-bit immediate value into the selected register. Neither instruction changes FLAGS.

ADD, SUB, AND, OR and XOR pass the selected register values to the ALU. The result is written into Register A, and Z, C and N are stored in FLAGS on the same rising edge.

CMP uses the ALU's subtraction operation and updates FLAGS without writing either register. Each instruction performs its register and flag updates once, then returns to instruction fetching.

Data-memory instructions

LOAD and IN read one byte from the address held in the Address Register. When the read completes, the byte is written into the selected register.

STORE and OUT write the selected register's byte to that address. They complete when memory acknowledges the write.

These instructions preserve FLAGS. IN and OUT use the same data-memory interface as LOAD and STORE; peripheral selection is handled by the memory map.

Jump instructions

JMP always loads the Address Register's target into the Program Counter.

JZ loads the target only when the stored Zero flag is one. JNZ loads it only when the stored Zero flag is zero. If the condition is false, the Program Counter retains the next sequential instruction address.

Jump instructions preserve registers and FLAGS.

Stack instructions

The stack grows towards lower RAM addresses.

PUSH calculates SP − 1 and writes the selected register's byte to that address. When the write completes, the Stack Pointer is updated to the calculated address.

POP reads the byte at the current Stack Pointer. When the read completes, the byte is written into the selected register and the Stack Pointer increments by one.

Both instructions preserve FLAGS. Stack addresses must remain within the configured stack region. A PUSH beyond its lower boundary or a POP from an empty stack causes a fault before any memory transaction or stored-value update occurs.

HALT and invalid instructions

HALT enters a stopped state. The CPU issues no further memory requests or register, flag, Program Counter or Stack Pointer updates until reset.

An unsupported opcode or nonzero reserved bit causes the CPU to enter a fault state and assert a fault indicator. The invalid instruction performs no execution actions. Reset clears the fault and restarts instruction fetching.

Control states

The control unit will use states for instruction fetch, decode, second-word fetch, execution, data-memory access, HALT and fault handling. Additional states may separate address preparation from memory transactions. Write enables default to zero and are asserted only for the cycle in which an update is required.

These rules define the intended behaviour. The next step is to implement and test the control unit before connecting it to the complete CPU.

**18. Control Unit Implementation and Testing**

ByteForge's clocked control unit was implemented in Verilog according to the execution rules in Section 17. It uses a finite-state machine to fetch instructions, decode them, fetch second words where required, and coordinate execution. It stores the Program Counter, Stack Pointer, Instruction Register, Address Register and memory-transaction information.

The control unit instantiates the previously tested instruction decoder. It generates timed register and FLAGS write enables, controls memory requests, and handles conditional jumps, PUSH, POP, HALT and faults. Memory requests and their payloads remain stable while waiting for acknowledgement. Updates occur once at the appropriate completion edge.

Stack boundaries are configurable. PUSH checks for available space before requesting a write, while POP checks that the stack is not empty before requesting a read. Invalid instructions and stack boundary violations enter a fault state. HALT and fault states remain stopped until reset.

A self-checking testbench was run using Icarus Verilog in EDA Playground. It completed 73 scenarios and 10,577 checks with zero failures. Testing covered instruction execution, register and flag updates, program addressing, jumps, memory delays, stack behaviour, invalid instructions, HALT, faults and reset during pending transactions.

This stage used behavioural models of the register file, ALU, FLAGS and memories. Integration with the actual previously tested CPU modules is the next stage.

The source files are control_unit.v and control_unit_tb.v, used alongside instruction_decoder.v. The simulation generated control_unit_waveform.vcd, and the results screenshot is saved as ByteForge_Control_Unit_Test_Results.png.

Saved simulation: ByteForge Control Unit Tests.